

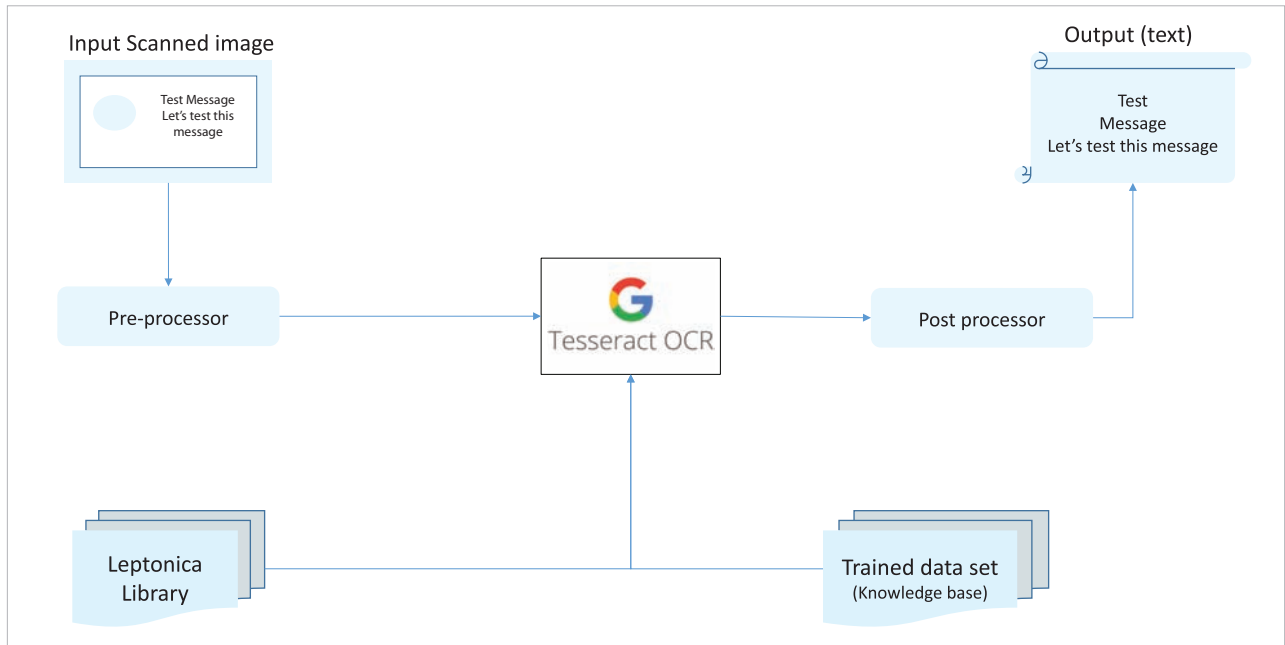


Figure 2: Execution process of Tesseract OCR

supports over 100 languages.

Another use case of deep learning is text summarization. Google found out that summarization can be done using a deep-learning technique called sequence-to-sequence (S2S) learning. Indeed, this S2S deep learning technique can be used to produce headlines for news articles. SmartReply is another Google use case, which automatically generates e-mail responses.

- c. **Image recognition:** Image recognition is used for face recognition, image search, motion detection, machine vision, and photo clustering. It can also be used in the automotive, aviation, and healthcare industries. The advantage of using TensorFlow for object recognition algorithms is that it helps to classify and identify arbitrary objects within larger images. This is used in engineering applications to identify shapes for modelling purposes (or 3D space reconstruction from 2D images), and by Facebook for photo tagging (Facebook's Deep Face). As an example, TensorFlow is used in deep learning for analysing

thousands of photos of dogs to help identify one.

- d. **Video detection:** Deep learning algorithms, these days, are also used for video detection like motion detection, real-time threat detection in gaming, for security at airports, and so on. For example, NASA is developing a deep learning system for orbit classification and object clustering of asteroids to classify and predict NEOs (near earth objects).

TensorFlow applications

The following are various TensorFlow applications developed by the open source community and used in diverse ways.

- a. **DeepSpeech:** DeepSpeech is a voice-to-text command and library, making it useful for those who need to transform voice input into text and developers who want to provide voice input for their applications. It is composed of two main subsystems — an acoustic model and a decoder. The acoustic model is a deep neural network that receives audio features as inputs, and outputs character probabilities. The decoder uses a beam search algorithm to

transform the character probabilities into textual transcripts that are then returned by the system.

- b. **RankBrain:** RankBrain is a component of Google's core algorithm based on TensorFlow, which uses machine learning to determine the most relevant results to search engine queries. Before RankBrain, Google utilised its basic algorithm to determine which results to show for a given query. Post-RankBrain, it is believed that the query now goes through an interpretation model that can apply possible factors like the location of the searcher, personalisation, and the words of the query to determine the searcher's true intent.
- c. **Inception v3:** Inception v3 is a pretrained convolutional neural network that is 48 layers deep, which is a version of the network already trained on more than a million images from the ImageNet database. This pretrained network can classify images into 1000 object categories, such as keyboard, mouse, pencil, and many animals. As a result, the network has learned rich